

PROJECT REPORT

STUDENT MANAGEMENT SYSTEM USING PYTHON

Python Essentials — Evaluated Course Project

Student Name	HARSH VARDHAN
Register / Admission No.	26BAI10807
Course	B.Tech CSE (AIML)
Project Type	Console-Based Python Application
Academic Purpose	Educational / Course Project

CERTIFICATE

This is to certify that HARSH VARDHAN, Register / Admission No. 26BAI10807, student of B.Tech CSE (AIML), has developed and documented the project titled "Student Management System using Python" as an educational Python project. The project demonstrates practical application of programming fundamentals, data handling, validation, exception handling, marks processing, and report generation.

DECLARATION

I, HARSH VARDHAN, Register / Admission No. 26BAI10807, declare that this project report documents the Student Management System using Python developed as an academic course project. The implementation and execution evidence presented are based on the supplied project materials.

ACKNOWLEDGEMENT

I would like to express my gratitude to my faculty and institution for providing the opportunity to develop this project. The project provided practical experience in Python programming, data structures, file handling, validation, exception handling, and structured problem solving.

ABSTRACT

The Student Management System is a Python-based console application designed to simplify the storage and management of student information and academic performance. It supports student record creation, viewing, searching, updating, deletion, marks entry, individual performance reporting, and consolidated performance reporting. Student data is persisted locally in JSON format. The implementation uses Python's standard library and includes validation, exception handling, duplicate-ID prevention, marks validation, grade calculation, pass/fail determination, and safe data-file replacement.

1. INTRODUCTION

Managing student records manually can become time-consuming and can make searching, updating, and maintaining academic information difficult. This project provides a compact computerized solution through a menu-driven Python application. The current implementation is console based and demonstrates core Python concepts in a practical academic context.

2. PROBLEM STATEMENT

Manual student record management may lead to duplicate records, difficulty in retrieval, and avoidable errors. The system addresses these issues through structured digital records and direct create, read, update, delete, marks, and reporting operations.

3. OBJECTIVES

Digitally manage student information; implement CRUD operations; store records using JSON; validate input; handle common errors; enter and update marks; calculate totals and percentages; assign grades; determine pass/fail; and generate individual and consolidated performance reports.

4. SCOPE AND TARGET USERS

The project covers student record management, searching, updating and deletion, marks management, total and percentage calculation, grade calculation, pass/fail determination, performance reports, JSON-based local storage, input validation, and error handling. Intended users include teachers, faculty members, academic administrators, students for basic viewing, and Python beginners.

5. TECHNOLOGIES AND REQUIREMENTS

Python 3; standard library; JSON; console interface. The README also documents Git, GitHub, and unittest as project tooling.

6. SYSTEM DESIGN AND WORKFLOW

At startup the program loads existing JSON data, displays the data-file location, shows the nine-option menu, executes the selected operation, saves changes where required, and returns to the menu until Exit is selected.

7. FUNCTIONAL MODULES

1 Add Student; 2 View All Students; 3 Search Student; 4 Update Student; 5 Delete Student; 6 Enter / Update Marks; 7 Generate Student Report; 8 Generate All Performance Reports; 9 Exit.

8. DATA STORAGE AND FILE HANDLING

Student data is stored as JSON in a writable application-data directory. The save routine writes to a temporary file, flushes and synchronizes it, and then replaces the existing JSON file. Loading handles missing files, invalid JSON, permission errors, and general exceptions.

9. VALIDATION AND ERROR HANDLING

Student IDs cannot be empty or contain spaces and duplicate IDs are rejected. Names cannot be empty and are restricted to alphabets and spaces. Courses cannot be empty. Marks must be numeric and within 0–100. Invalid menu choices are rejected.

10. MARKS AND PERFORMANCE LOGIC

The four subjects are Python, Mathematics, Computer Fundamentals, and Communication Skills. Grade thresholds are A at 90+, B at 75+, C at 60+, D at 50+, otherwise F. A student passes only when every entered subject mark is at least 40. Percentage is calculated from total marks and the maximum possible marks.

11. PROGRAM STRUCTURE AND FUNCTIONS

The source contains dedicated functions for data location, loading/saving, validation, marks calculation, student CRUD operations, marks entry, individual reports, consolidated reports, menu display, and the main control loop.

12. TESTING AND EXECUTION RESULTS

The supplied screenshots demonstrate adding, viewing, searching, updating, deleting, entering marks, generating a student report, generating all performance reports, and exiting. The observed sample workflow completed successfully.

14. SAMPLE PERFORMANCE REPORT

Student ID / Register No.: 26BAI10807; Name: HARSH VARDHAN; Course: B.Tech CSE (AIML); Semester: 1st SEMESTER. Marks: Python 99, Mathematics 90, Computer Fundamentals 95, Communication Skills 100. Total: 384/400; Percentage: 96.00%; Grade: A; Result: Pass; Performance: Excellent Performance.

15. LEARNING OUTCOMES

The project demonstrates variables, data types, operators, conditions, loops, functions, dictionaries, file handling, JSON, exception handling, validation, modular programming, testing, Git, and GitHub.

16. LIMITATIONS

Console-based interface; local JSON rather than a relational database; no authentication or role management; no PDF export in the current implementation; predefined subject list; no graphical dashboard.

17. FUTURE ENHANCEMENTS

The supplied README proposes Tkinter GUI, login and role management, SQLite, attendance management, PDF report export, data visualization, and cloud storage.

18. CONCLUSION

The Student Management System demonstrates how fundamental Python concepts can be combined into a practical record-management application. It provides a complete menu-driven workflow, persistent JSON storage, validation, error handling, and readable academic performance reports.

19. REFERENCES

Project Statement; README; Source Code; Final VITYarthi project.py; execution screenshots supplied with the project.

13. EXECUTION SCREENSHOTS

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Enter / Update Marks
7. Generate Student Report
8. Generate All Performance Reports
9. Exit
=====
```

Figure 1. Main Menu

```
=====
Enter your choice (1-9): 1
=====
                        ADD STUDENT
=====
Enter Student ID: 26BAI10807
Enter Student Name: HARSH VARDHAN
Enter Course: B.Tech CSE (AIML)
Enter Year/Semester: 1st SEMESTER

-----
Student added successfully!
-----

Student ID : 26BAI10807
Name       : HARSH VARDHAN
Course     : B.Tech CSE (AIML)
Year       : 1st SEMESTER
```

Figure 2. Add Student

Enter your choice (1-9): 2

ALL STUDENTS

ID	Name	Course	Year
26BAI10807	HARSH VARDHAN	B.Tech CSE (AIML)	1st SEMESTER

Figure 3. View All Students

```
=====
Enter your choice (1-9): 3

=====
                        SEARCH STUDENT
=====
Enter Student ID or Name: HARSH VARDHAN

Student Found
-----
Student ID : 26BAI10807
Name       : HARSH VARDHAN
Course     : B.Tech CSE (AIML)
Year       : 1st SEMESTER
```

Figure 4. Search Student

```
=====
Enter your choice (1-9): 4

=====
                        UPDATE STUDENT
=====
Enter Student ID: 26BAI10807

Press ENTER to keep the existing value.
Enter new name [HARSH VARDHAN]: Harsh Vardhan
Enter new course [B.Tech CSE (AIML)]: AIML
Enter new year/semester [1st SEMESTER]: 1st

Student updated successfully!
```

Figure 5. Update Student

```
=====
Enter your choice (1-9): 5
=====
```

```
=====
                        DELETE STUDENT
=====
```

```
Enter Student ID: 26BAI10807
```

```
Student Details
=====
```

```
Student ID : 26BAI10807
Name       : Harsh Vardhan
Course     : AIML
```

```
Are you sure you want to delete this student? (y/n): y
```

```
Student deleted successfully!
```

Figure 6. Delete Student

```
=====
Enter your choice (1-9): 6
=====
```

```
=====
ENTER / UPDATE MARKS
=====
```

```
Enter Student ID: 26BAI10807
```

```
Student: HARSH VARDHAN
=====
```

```
Enter marks for Python (0-100): 99
```

```
Enter marks for Mathematics (0-100): 90
```

```
Enter marks for Computer Fundamentals (0-100): 95
```

```
Enter marks for Communication Skills (0-100): 100
```

```
Marks saved successfully!
```

Figure 7. Enter / Update Marks

```
=====
Enter your choice (1-9): 7
=====
```

```
=====
STUDENT REPORT
=====
```

```
Enter Student ID: 26BAI10807
```

```
=====
STUDENT PERFORMANCE REPORT
=====
```

```
Student ID : 26BAI10807
Name       : HARSH VARDHAN
Course    : B.Tech CSE (AIML)
Year      : 1st SEMESTER
```

```
Marks
=====
```

```
Python           : 99.00
Mathematics      : 90.00
Computer Fundamentals : 95.00
Communication Skills : 100.00
=====
```

```
Total Marks : 384.00
Percentage  : 96.00%
Grade      : A
Result     : Pass
Performance : Excellent Performance
=====
```

Figure 8. Student Performance Report

```
=====
Enter your choice (1-9): 8

=====
                        ALL PERFORMANCE REPORTS
=====
ID           Name           Percentage   Grade   Result
=====
26BAI10807  HARSH VARDHAN           96.00     A       Pass
=====
```

Figure 9. All Performance Reports

```
=====
                        STUDENT MANAGEMENT SYSTEM
=====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Enter / Update Marks
7. Generate Student Report
8. Generate All Performance Reports
9. Exit
=====
Enter your choice (1-9): 9

=====
Thank you for using Student Management System!
Goodbye!
=====
harshvardhan@HARSHs-MacBook-Air / % █
```

Figure 10. Program Exit